

DM2026 Assignment 3: Human Activity Recognition

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GitHub Repository: <https://github.com/RaisoLiu/DM2026-Assignment-3>

Kaggle Public Score: 0.8106, public leaderboard, submitted 2026-05-05.

Selected Submission File: submissions/submission_lgbm_leaves63_calibrated.csv

Selected Submission SHA-256:

f78034a7e72d25ab4d52baebed585e130ad810e1f5cfaf7469205008d5d5035e

1 How to Run the Code

The implementation uses Python 3 and the packages listed in `requirements.txt`. The Kaggle archive is downloaded with:

```
kaggle competitions download -c nycu-data-mining-assignment-3 -p data/raw --force
unzip -q -o data/raw/nycu-data-mining-assignment-3.zip -d data/raw
```

The downloaded archive contains `train/train/User_*` and `test/test/User_*`. I move the `User_*` directories one level up so the final layout is:

```
data/raw/train/User_001/*.csv ... data/raw/train/User_060/*.csv
data/raw/test/User_061/*.csv ... data/raw/test/User_100/*.csv
data/raw/sample_submission.csv
```

The final Kaggle CSV is reproduced by:

```
python scripts/make_boosting_submission.py \
--results-csv artifacts/model_search_position_proba/results_position.csv \
--model lgbm_leaves63 \
--context position \
--output submissions/submission_lgbm_leaves63_calibrated.csv \
--seed 2026
```

`make_boosting_submission.py` trains the explored LightGBM model on all labeled windows, applies the OOF class calibration weights, and writes a Kaggle-compatible `Id,Label` CSV aligned to `sample_submission.csv`. The earlier HGB baseline and feature-family ablations are reproduced by `scripts/run_experiment.py`.

2 Preliminary Analysis and Design Motivation

The training set contains 11,020 five-minute windows from 60 users. The test set contains 6,849 windows from 40 disjoint users. Each file has 300 one-second rows with `mean_x`, `mean_y`, `mean_z`, `std_x`, `std_y`, `std_z`, `index`, and `file_id`; training files also contain one repeated file-level `label`.

The task is strongly imbalanced (Figure 1). Since the official metric is macro-F1, accuracy is not sufficient. A model can achieve high accuracy by mostly predicting labels 0 and 1, but it will lose macro-F1 on the minority activities. I therefore use user-grouped cross-validation and class-balanced training weights. Grouping by `User_*` is important because the test users are disjoint from training users; random window splits would leak user-specific orientation, calibration, and wearing-style information.

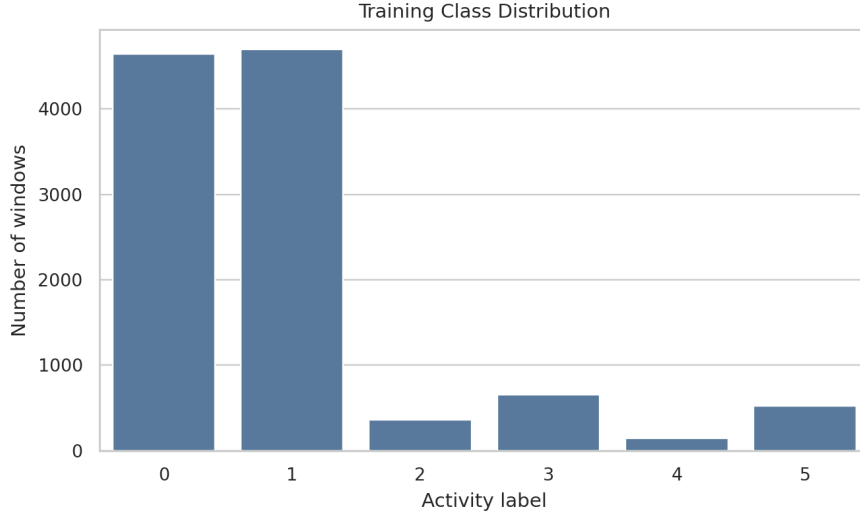


Figure 1: Training class distribution. Labels 0 and 1 account for 84.7% of training windows, while label 4 has only 142 examples.

Table 1: Training class counts and fractions

Label	Count	Fraction
0	4,643	42.13%
1	4,695	42.60%
2	358	3.25%
3	656	5.95%
4	142	1.29%
5	526	4.77%

The accelerometer retains the gravity component. This makes vector magnitude and orientation informative: activities can differ by both motion intensity and wrist posture. The preliminary feature check showed that the average mean-vector magnitude is close to 1g (`mean_vm_mean` median 0.997), while dynamic variability (`std_vm_mean`) ranges widely. This motivated a file-level feature representation combining static gravity-aware features and temporal variability features.

3 Preprocessing and Feature Engineering

Each CSV is sorted by `index`, interpolated within the six signal channels, and converted into one feature vector. I do not train on individual seconds as independent samples because the label belongs to the whole five-minute window.

The base feature table has 399 engineered features per window. The final LightGBM submission adds six user-sequence position features, giving 405 features:

Table 2: Feature families used by the selected LightGBM model

Feature family	Purpose
Global channel statistics	Quantiles, RMS, range, skewness, and kurtosis.
Gravity/magnitude	Vector magnitude, deviation from 1g, dynamic/static ratio.
Temporal derivatives	First-difference volatility, energy, and zero-crossings.
Segment/rolling	Five one-minute segments plus 5/15/30-second rolling variability.
Frequency	FFT power, dominant frequency, entropy, centroid, and band ratios.
Cross-axis/orientation	Axis-normalized orientation, correlations, and covariance.
User sequence position	Normalized order, reverse order, edge distance, sine/cosine phase, and sequence length.

The most important preprocessing choice is keeping the prediction unit at the file level. Feature extraction and final submission both align by `file_id`; the submission is merged back to `sample_submission.csv` by `Id`.

4 Label Alignment and Temporal Dependency Modeling

Each 300-row file corresponds to exactly one activity label. The pipeline uses the following alignment:

1. Read one CSV file and sort rows by `index`.
2. Infer the single `file_id` and, for training, the single label.
3. Extract one window-level feature vector from all 300 seconds.
4. Train and validate on one row per file.
5. Predict one label per test `file_id`.
6. Merge predictions to `sample_submission.csv` in the original `Id` order.

Temporal dependencies are represented through deterministic sequence summaries. First differences capture second-to-second motion changes. Segment statistics preserve coarse order over the five-minute interval. Rolling statistics capture local bursts or sustained variability. FFT summaries capture periodicity at the available 1 Hz resolution. This design is simpler and more reproducible than a deep sequence model, while still respecting the temporal nature of the data.

5 Core Model and Validation Strategy

The final submitted model is `LGBMClassifier` with `num_leaves=63`, 900 estimators, class-balanced sample weights, and the 405-feature position-augmented window representation. I kept the HGB pipeline as the baseline for feature-family ablations because it is deterministic, fast, and easy to stress-test. The validation protocol is 5-fold grouped cross-validation by user with seed 2026. Grouped CV is deliberately more conservative than a random window split because the public/private test set contains unseen users.

The selected prediction applies out-of-fold class-bias calibration. I first collect OOF predicted probabilities from grouped CV, then tune one multiplicative decision weight per class to maximize macro-F1 on the OOF predictions. This does not retrain the model or use test labels; it only adjusts the decision rule to match the official macro-F1 objective.

Table 3: Class decision weights selected from OOF macro-F1 calibration

Class	Weight
0	0.41647
1	0.45553
2	5.85992
3	0.89605
4	1.31184
5	0.76524

For the final LightGBM model, calibration increased grouped OOF macro-F1 from 0.72509 to 0.74771. The main gain was improved recall for label 2, a rare class that otherwise tends to be absorbed into labels 1 and 3.

Table 4: Grouped CV and public leaderboard model search summary

Model / decision rule	OOF Macro-F1	Public score
HGB baseline, calibrated	0.73823	0.7922
XGBoost position model, calibrated	0.74977	0.7933
LightGBM leaves63 position model, calibrated	0.74771	0.8106

Table 5: Calibrated 5-fold grouped CV classification report

Label	Precision	Recall	F1
0	0.97537	0.98083	0.97809
1	0.90484	0.91949	0.91211
2	0.28536	0.32123	0.30223
3	0.68714	0.73323	0.70944
4	0.92537	0.87324	0.89855
5	0.86881	0.56654	0.68585
Macro avg	0.77448	0.73243	0.74771

LGBM calibrated OOF confusion matrix

True label	0	1	2	3	4	5
0	4554	71	0	16	0	2
1	112	4317	158	80	2	26
2	2	175	115	59	0	7
3	1	84	72	481	8	10
4	0	2	1	15	124	0
5	0	122	57	49	0	298
	0	1	2	3	4	5
	Predicted label					

Figure 2: Calibrated LightGBM 5-fold OOF confusion matrix. The dominant residual error is label 2 being confused with labels 1 and 3.

6 Ablation Study

I used the same fast HGB configuration for feature-family ablations. Each ablation removes one feature family and reruns 5-fold grouped CV.

Table 6: Feature-family ablation with grouped CV

Design	Macro-F1	Delta from full
Full feature set	0.72921	0.00000
Remove gravity/magnitude	0.71548	-0.01373
Remove temporal derivatives	0.72808	-0.00113
Remove segment/rolling	0.73131	+0.00210
Remove frequency	0.72923	+0.00002
Remove orientation/cross-axis	0.73168	+0.00246

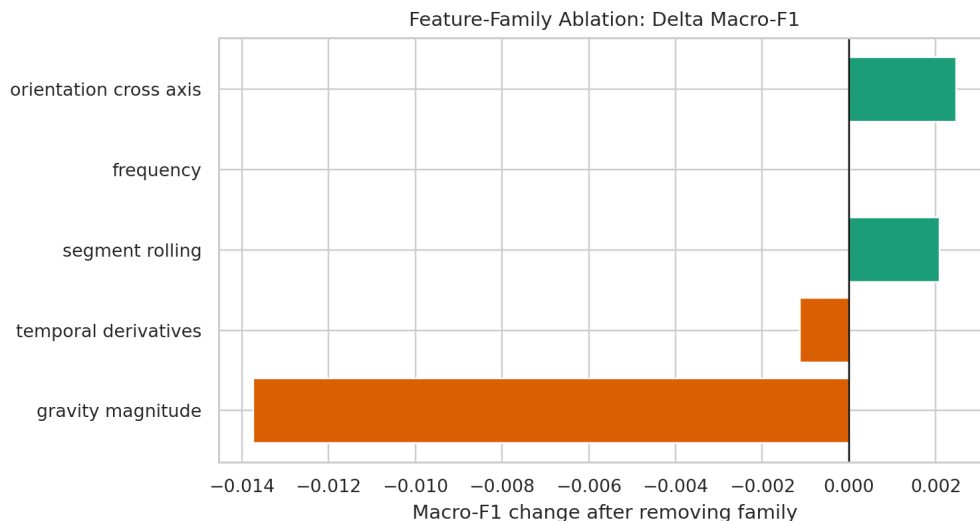


Figure 3: Macro-F1 change after removing each feature family. Gravity/magnitude features are the clearest positive contributor.

The largest loss comes from removing gravity/magnitude features, confirming that the retained gravity component is useful. Removing segment/rolling or orientation/cross-axis features slightly improves the uncalibrated HGB score in this split, suggesting mild fold-specific noise. I therefore used the ablation as evidence for the feature representation rather than as a final model selector. The final LightGBM submission keeps the complete engineered feature set and adds only compact user-position features.

I also tested a full 450-iteration HGB. It reached calibrated OOF macro-F1 0.74751, but its calibrated test predictions assigned label 2 to 15.56% of test windows, compared with only 3.25% in the training set. The final LightGBM submission has a more plausible test prior: label 2 is 2.76%, label 4 is 1.26%, and labels 0/1 remain dominant. It also produced the best observed public leaderboard score, 0.8106 (Figure 4).

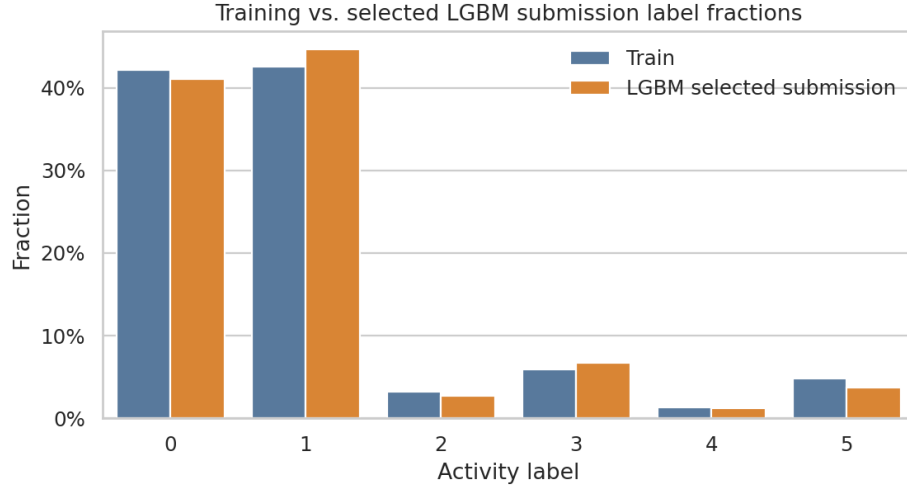


Figure 4: Training label fractions compared with the final LightGBM submission’s predicted label fractions. The selected submission keeps rare-class priors close to the training distribution while improving public leaderboard score.

7 Residual Risks and Next Steps

The main residual error is label 2. Even after calibration, label 2 F1 is only 0.30223 in grouped OOF validation, with many examples confused as labels 1 or 3. This is the highest-value target for future improvement because macro-F1 weights every class equally. The next experiments I would prioritize are:

- stronger label-2 focused calibration constrained by plausible test priors;
- user-level bootstrap stability checks for calibration weights;
- a small sequence model or 1D convolution benchmark using the same grouped validation protocol;
- ensembling only models whose test label priors remain plausible.

The final selected CSV was submitted to Kaggle and scored 0.8106 on the public leaderboard. The submitted file is format-checked: 6,849 rows, columns exactly `Id,Label1`, label range 0–5, no missing labels, and ID order matching `sample_submission.csv`. I keep all submitted files and scores in `SUBMISSION_LOG.md`; the competition allows three submissions per day.